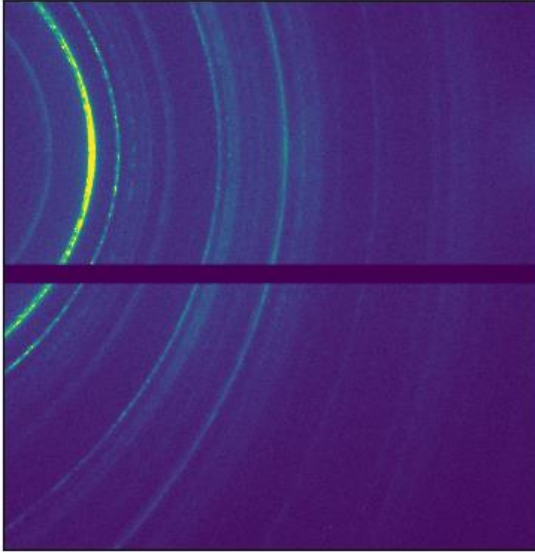


What You Will Learn from This Tutorial

- Conventional way of doing phase identification
- Basic machine learning workflow for phase identification
- Technique reviews
 - Mixtures
 - Probabilistic labeling
 - Interpretability
 - End-to-end structure prediction
- Underexplored topics
 - Novelty detection
 - Probability calibration
 - Texturing
- Problems we are facing
 - Benchmark dataset
 - Inductive bias

What Contributes to a Phase in a XRD Spectrum?

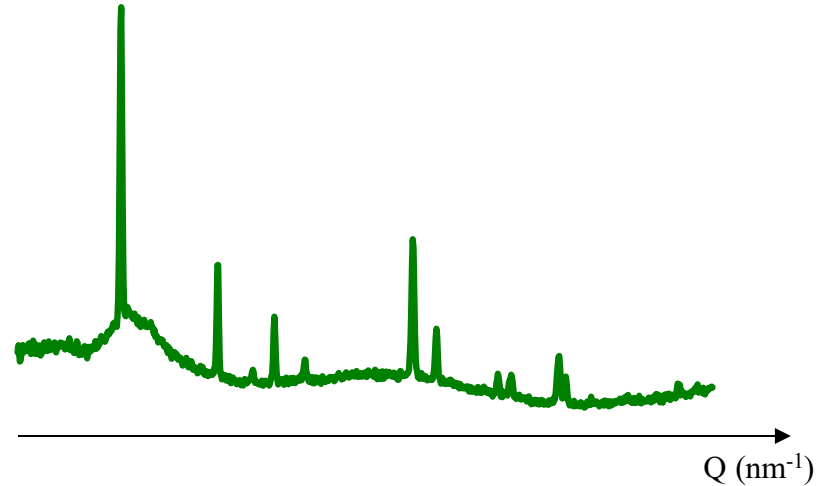
- 2D



- Rings
- Structure in the ring

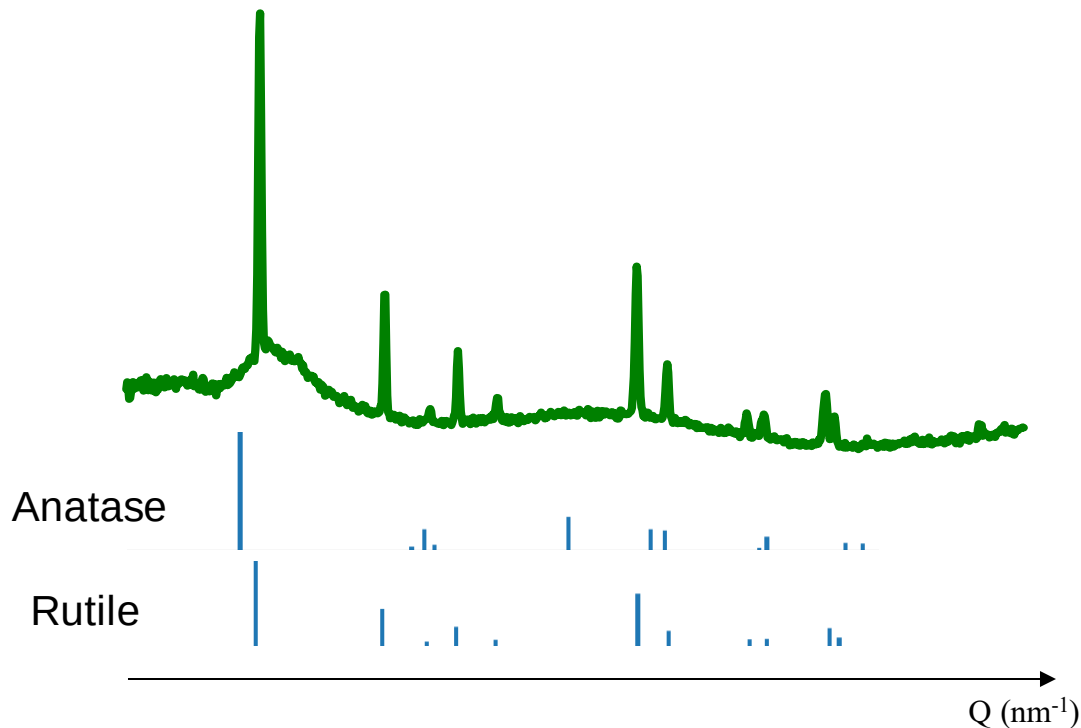
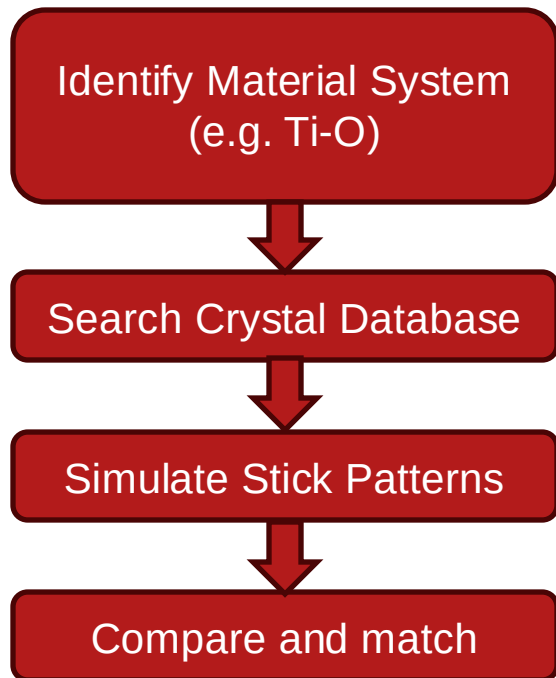
- 1D

Azimuthal
Integration

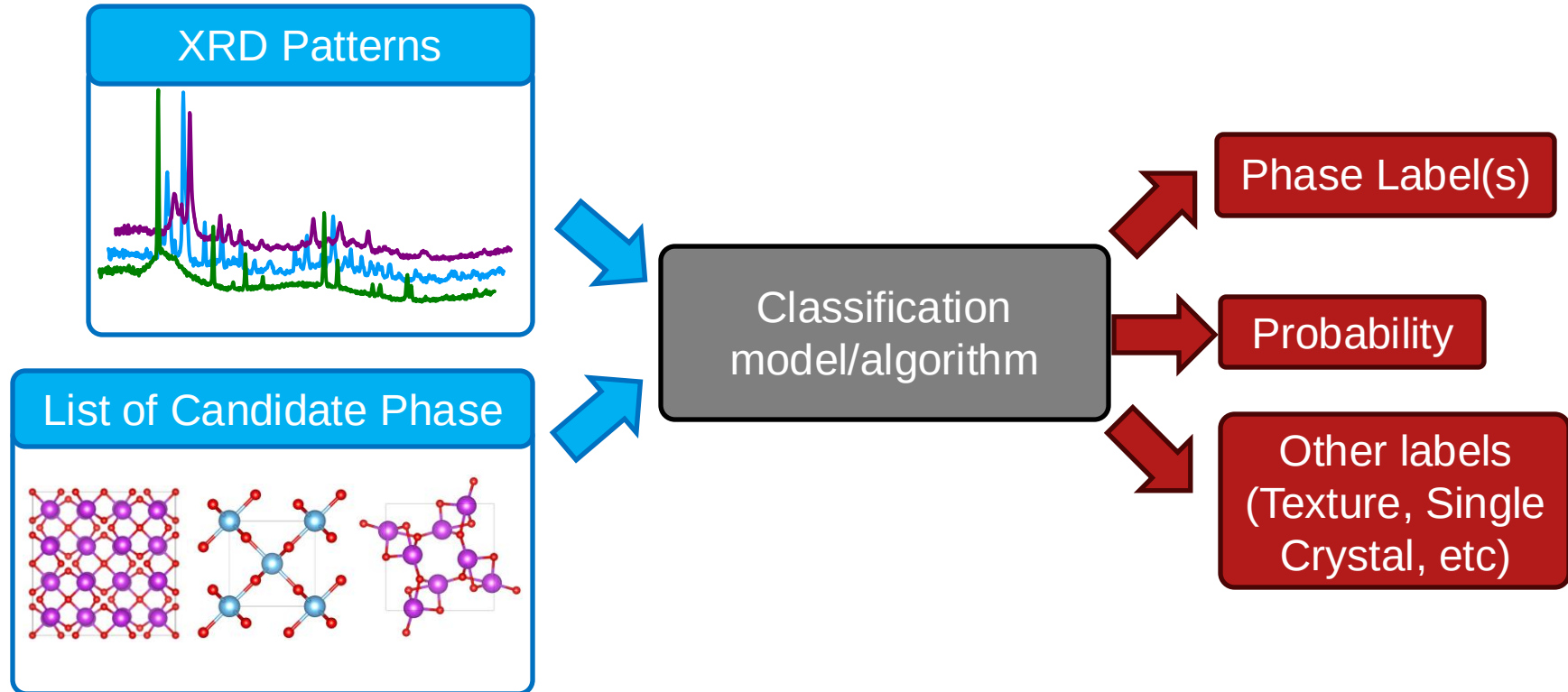


- Peak position
- Peak intensity

Manual Phase Identification

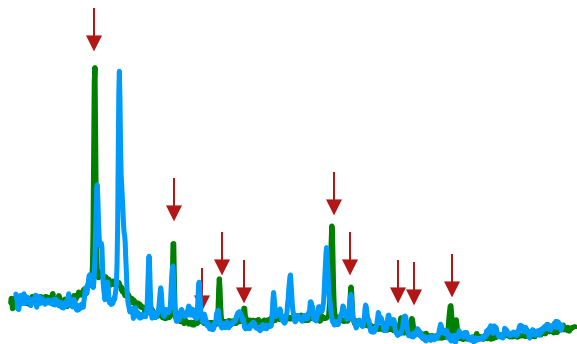


Automatic Phase Identification: Problem Definition



Automatic Phase Identification

Peak detection



- Use peak detection algorithm to find peak positions
- Compare to those calculated in a database
- Some choosing criteria (correlation, cosine similarity, etc.)
- Problem: when you have overlapping peaks or if you have more than one phase

Full pattern matching

1. Correlation

- Compare data with simulated XRD pattern
- Difficult to select criteria

2. Refinement

Methods

1. Rietveld Refinement: Refine crystal structure based on XRD
 2. Pawley Method: Refine peak position and fit peak intensity
 3. Le Bail Method: Iterating between cell refinement and refine peak intensity
- Use goodness-of-fit (e.g. R_{wp} , R_p) to assign phase
 - Time-consuming and laborious

3. Refinement + Search

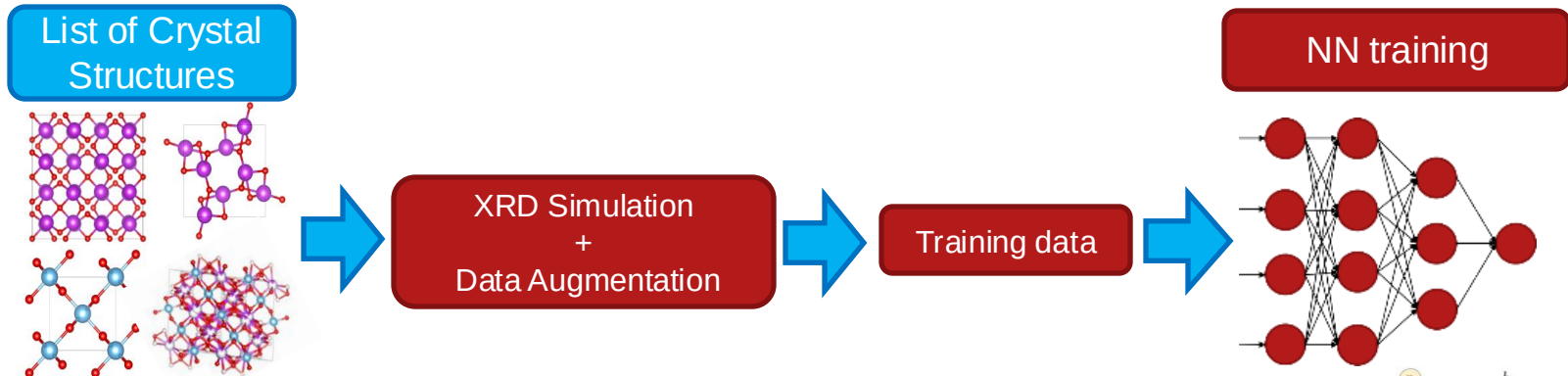
- Refinement of potential candidate phase
- Use some sort of search to deal with multiphase case
- Construct decision model based on the above result
- e.g. CrystalShift ([arXiv:2308.07897](https://arxiv.org/abs/2308.07897))
- Still hit or miss

Why Machine Learning?

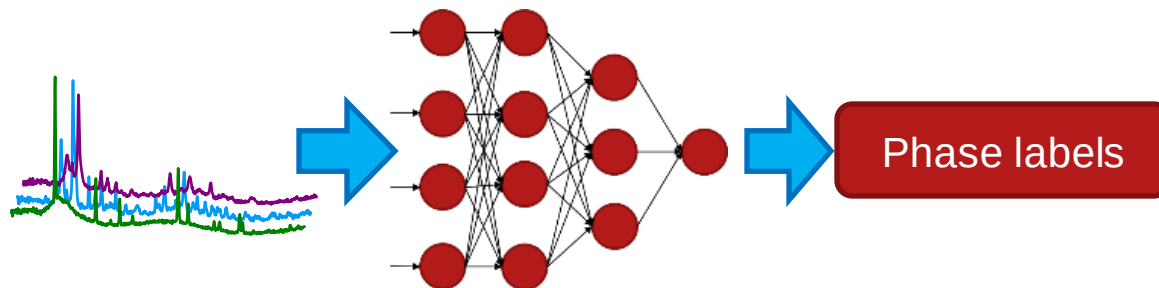
1. Replace human intuition and experience with training dataset
2. Very fast at inference time
3. ****More Accurate****
4. Core of phase identification problem is pattern recognition and matching
 - Similar to tasks ML are good at, e.g. computer vision, natural language processing
 - A lot of readily developed method and framework
5. Potential of integrating with other aspect of materials
 - E.g. direct integration with formation energy predictions
 - Multimodal models

ML Process for Phase Identification

Training

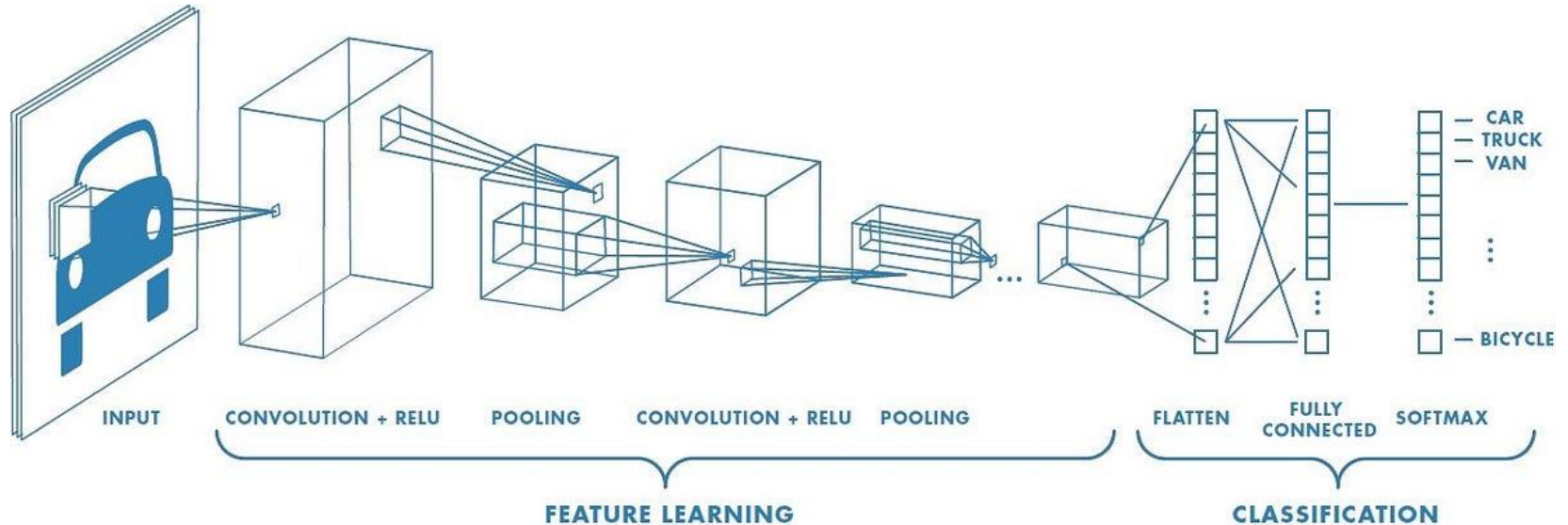


Inference

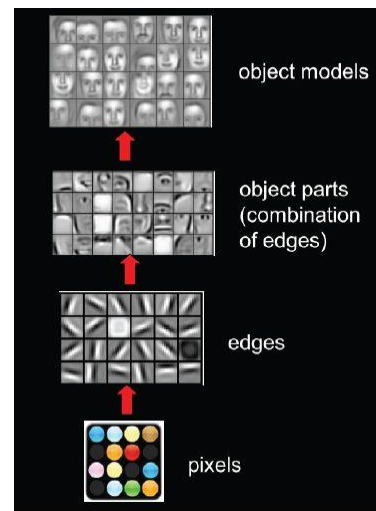
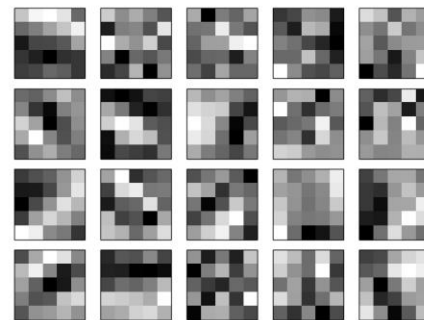
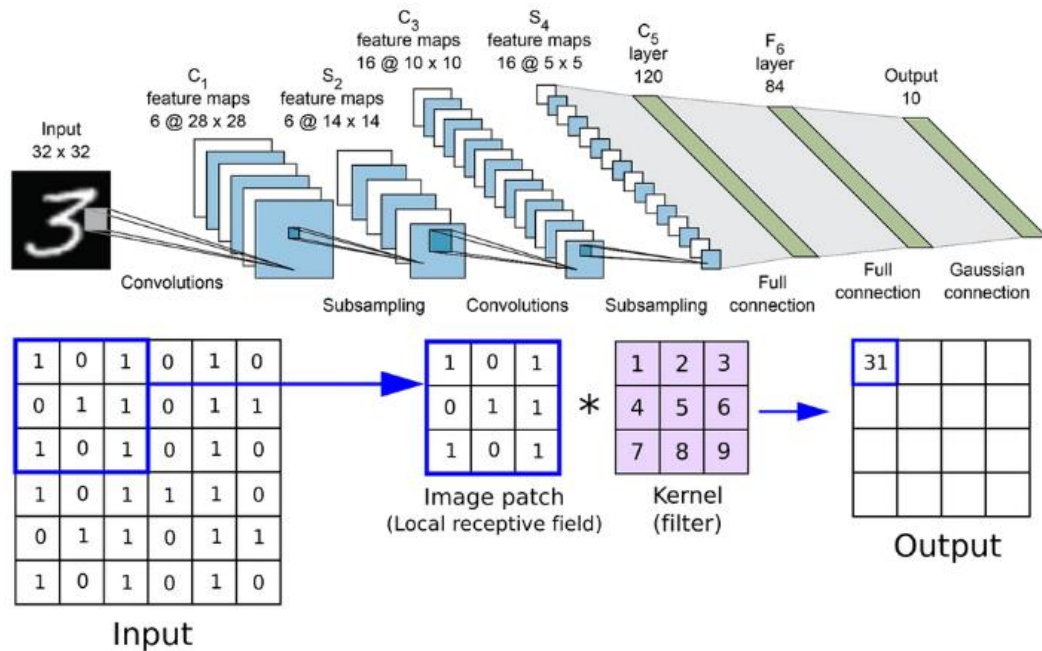


$$\begin{matrix} x_1 & w_1 \\ x_2 & w_2 \\ \vdots & \vdots \\ x_n & w_n \end{matrix} \rightarrow \Sigma \rightarrow f \left(b + \sum_{i=1}^n x_i w_i \right)$$

Convolutional Neural Network (CNN)



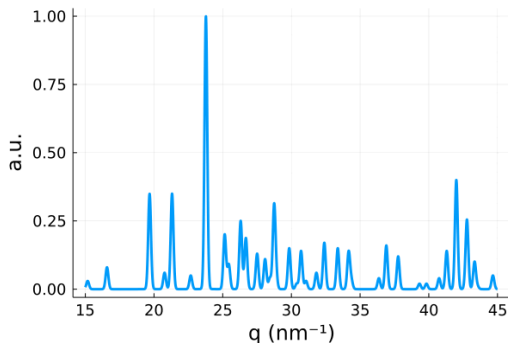
Convolutional Neural Network (CNN)



Data Augmentation

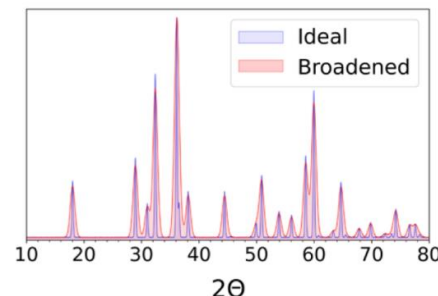
- **Purpose:** Provide more diverse training data and to prevent overfitting

Peak Shifting



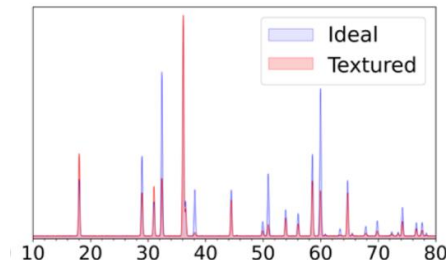
- Strained crystal leads to peak shifting
- Symmetry consideration

Peak Broadening



- Simulates
 - strain broadening
 - Scherrer broadening
 - apparatus broadening

Peak Ratio Modification



- Texturing: Preferred crystal orientation
- Single Crystal: may even eliminate some peaks
- Atomic Disorder: change peak ratio and cause peak to shift

Others

Peak shape: Voigt profile parameters

Background:

1. Gaussian background
2. Polynomial background

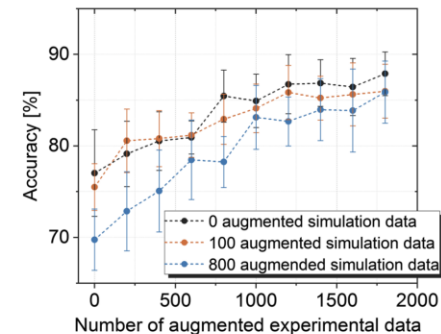
Performance on Vanilla Implementations

Table 1. Results of the ML algorithms under consideration

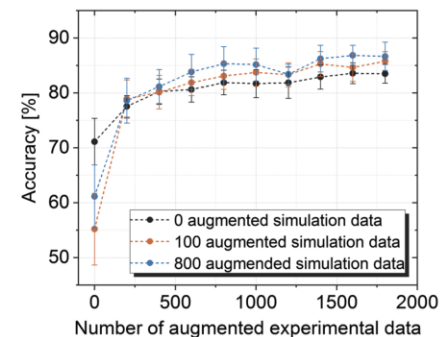
Method	Case 1 accuracy [%]	Case 2 accuracy [%]	Case 3 accuracy [%]	Case 3 micro F ₁ score	Case 3 macro F ₁ score
Naive Bayes	84.9/81.2	66.7/25.5	83.3/60.5	0.833/0.605	0.830/0.433
K-nearest neighbors	95.1/93.2	68.3/21.2	83.3/74.5	0.833/0.745	0.820/0.621
K-nearest neighbors with dynamic time warping	96.0/94.1	75.3/78.5	86.6/80.1	0.866/0.801	0.836/0.645
Logistic regression	92/82.3	68.3/13.9	91.7/76.6	0.917/0.766	0.847/0.601
Random forest	93.8/81.6	63.3/56.2	83.3/82.1	0.833/0.821	0.827/0.597
Decision tree	87.1/80.7	45.2/31.2	81.7/69.1	0.817/0.691	0.793/0.550
Support vector machine	94.4/93.7	67.1/67.3	88.1/78.5	0.881/0.785	0.805/0.659
Gradient boosting regression trees	96.2/92.1	62.1/62.1	89.7/81.1	0.897/0.811	0.866/0.653
Dense neural network	99.2/98.1	84/75.4	92.1/84.1	0.921/0.841	0.858/0.695
All convolutional neural network (a-CNN)	99.1/99.1	83.7/80.2	92.9/89.3	0.929/0.893	0.859/0.714

Reported average accuracy, *micro* and *macro* F₁ score for dimensionality/space group classification after fivefold cross-validation for each classification algorithm and case. In Case 1, exclusively simulated XRD patterns are used for testing and training. In Case 2, the simulated XRD patterns are used for training, and the experimental patterns for known materials are used for testing. In Case 3, all of the simulated data and 80% of the experimental data are used for training, and 20% of the experimental data are used for testing. The best performing algorithm is our a-CNN architecture

(a) Dimensionality



(b) Space group

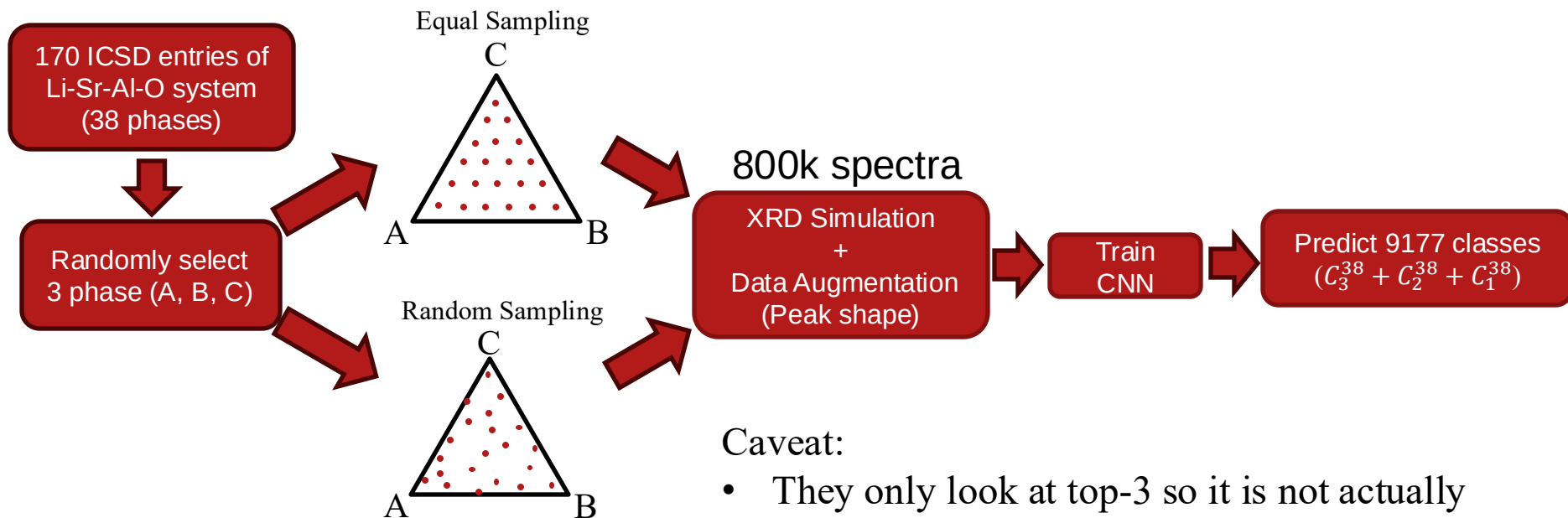


Dealing with Mixtures

- It is common to have phase coexistence in samples
- Overlapping of peaks and complicated peak shifting makes decomposition very difficult
- Combinatorial problem
 - Complexity $O(n^d)$ with n phase and up to d # phase coexistence
- Probabilistic labeling preferred

Brute Force

- It grows exponentially, so be it

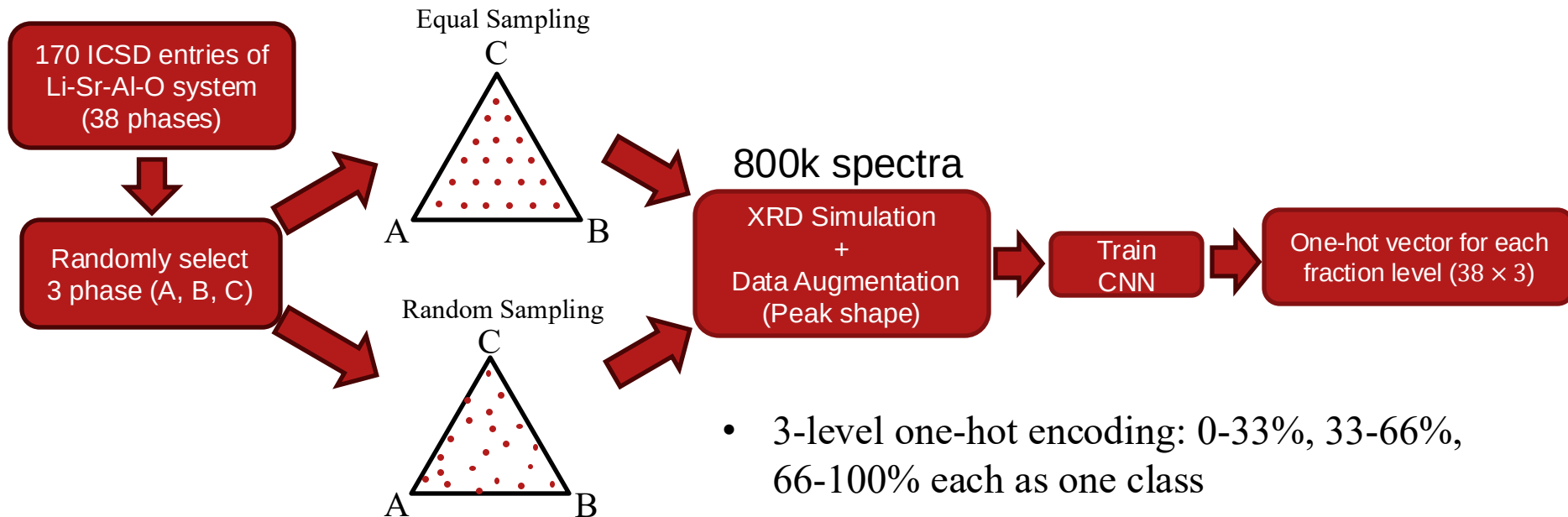


Caveat:

- They only look at top-3 so it is not actually choosing how many phases

Brute Force

- Fraction Prediction



Brute Force

Table 1 Phase identification test result.

Dataset	Dataset_800k_Org		Dataset_800k_Rand	Dataset_180k_Rand
	CNN_2	CNN_3	CNN_3	CNN_3
Simulated XRD test dataset				(23,000 patterns)
100,000 Patterns	99.60%	100%	100%	99.76%
Real XRD test dataset				
Li ₂ O_SrO-Al ₂ O ₃ (50 patterns)	100%	100%	100%	98.67%
SrAl ₂ O ₄ _SrO-Al ₂ O ₃ (50 patterns)	97.33%	98.67%	98%	97.33%

The phase identification test results from both the simulated (hold-out) and real test datasets for CNN_2 and CNN_3 trained with Dataset_800k_org, Dataset_800k_rand, and Dataset_180k_rand. The CNNs were trained for two epochs

Table 2 Three-step-phase-fraction prediction test result.

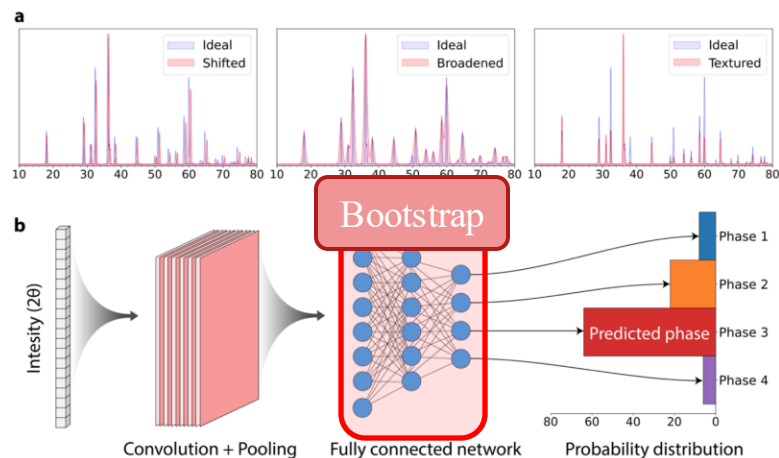
CNN Architecture	CNN_2F	CNN_3F	CNN_4F	CNN_5F	CNN_6F
Simulated XRD test dataset (Dataset_800k_Org)					
100,000 Patterns	94.39%	97.19%	97.64%	98.08%	98.13%
Real XRD test dataset					
Li ₂ O_SrO-Al ₂ O ₃ (50 patterns)	76.00%	80.00%	83.33%	86.00%	82.67%

The three-level-phase-fraction prediction test results based on Dataset_800k_org. The CNNs were trained for 10 epochs

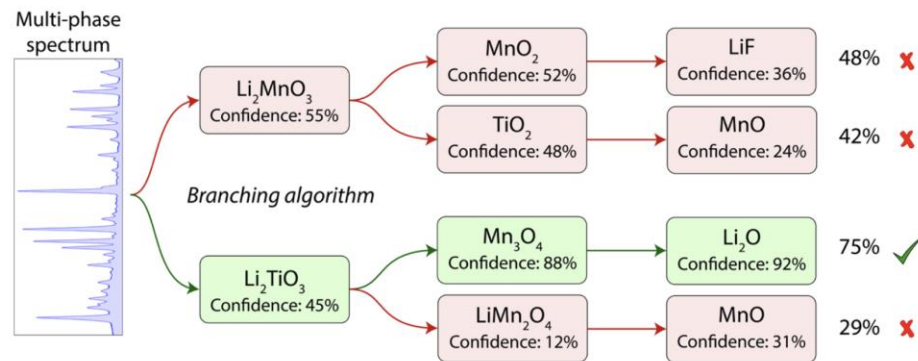
- Very sketchy results
 - Ultrahigh training and testing accuracy
 - Unable to reproduce because the data they provide is unreadable
- Heavily biased training data
 - 74 Al₂O₃, 14 Li₂O and 7 SrO repeated entries
 - Although they remove the sample if they all are selecting the same phase, the overrepresented phase are still sampled more
- No actual mechanism of choosing number of phases

Search with ML

- To circumvent the combinatorial problem, use detect and deduct approach



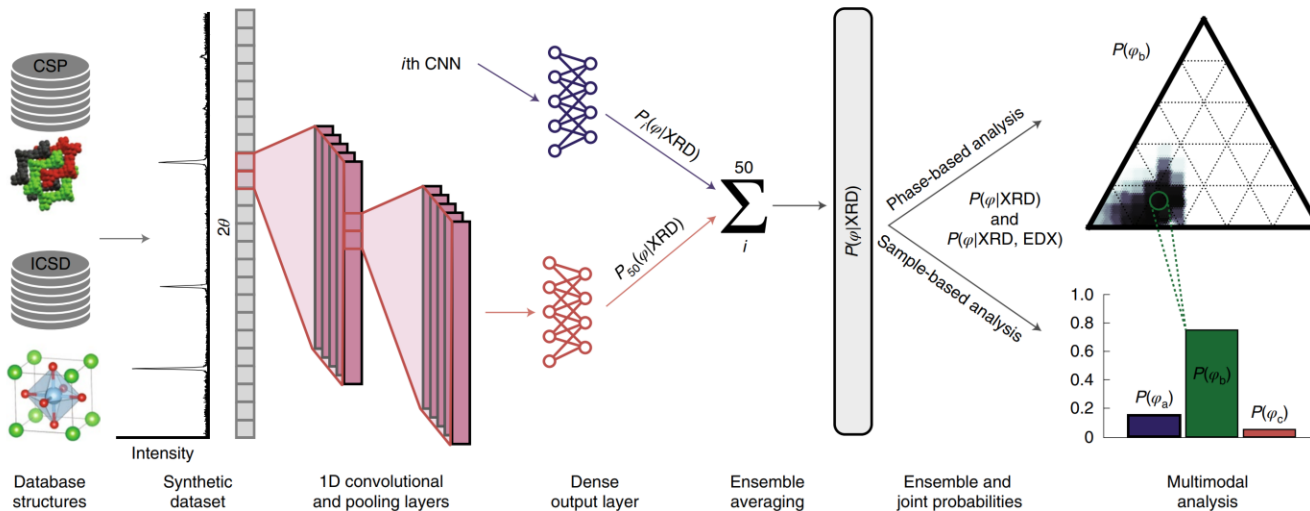
- Training data augmented by homogeneous strain, broadening, texturing, background noise
- Use dropout to sample the trained NN to gauge confident level
- Confident level of phase₁ = $\frac{\# \text{ of phase}_1 \text{ predicted}}{\# \text{ of times sampled}}$



- Once a phase is detected, dynamic time warping (DTW) is used to shift the simulated XRD to best match the given pattern
- Repeat until < 5% signal left
- Choose branch with highest average confidence

Ensemble Model for Probability Estimation

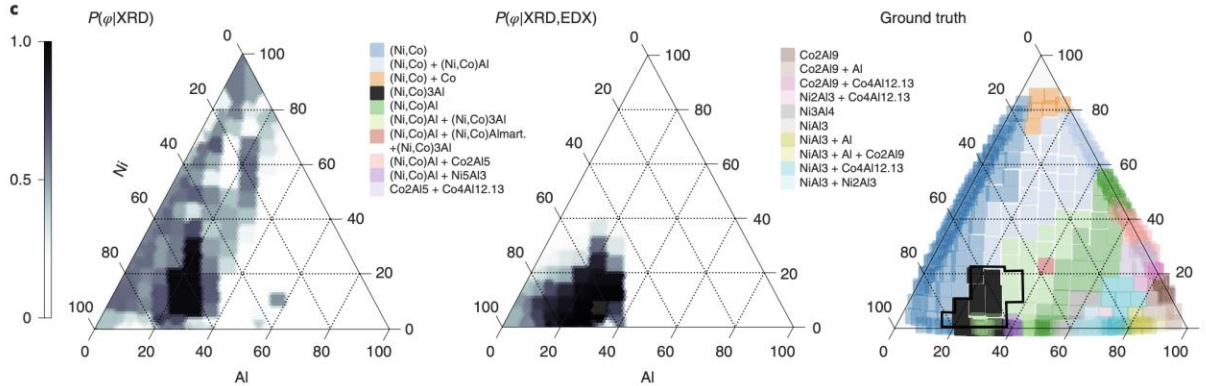
- Crystallography Companion Agent



- Train many small, shallow neural network and let them vote
- Use the fraction of vote as confident level

Ensemble Model for Probability Estimation

- Crystallography Companion Agent
 - Use composition data to assist phase assignment



$$P(\varphi_i|XRD) = \frac{1}{n} \sum_i^n f_n(\varphi_i)$$

f_n : nth shallow neural network

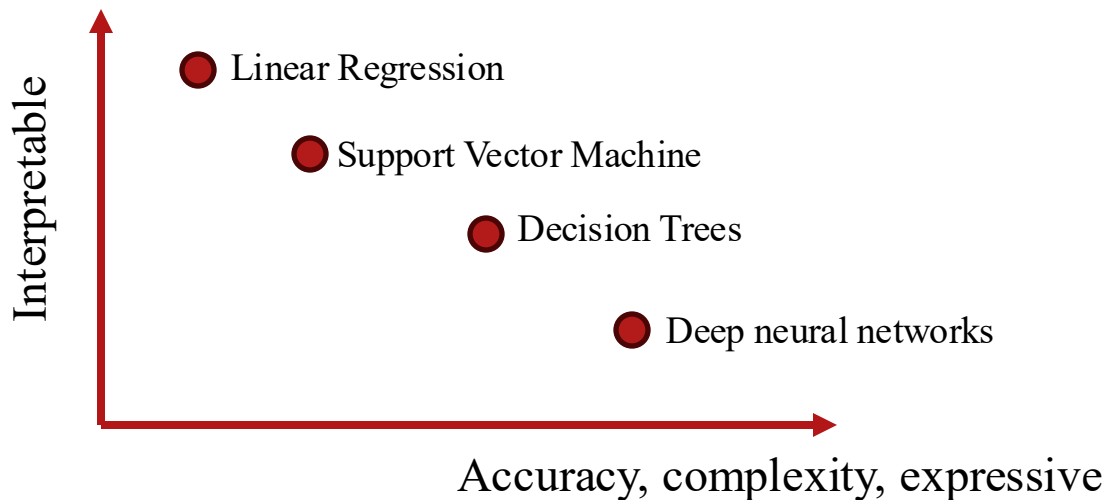
$$P(\varphi_i|EDX) = \prod_a \exp \frac{(x_a - x_a^i)^2}{\sigma^2}$$

$$\frac{P(\varphi_i|XRD, EDX)}{\sum_i P(\varphi_i|XRD) P(\varphi_i|EDX)}$$

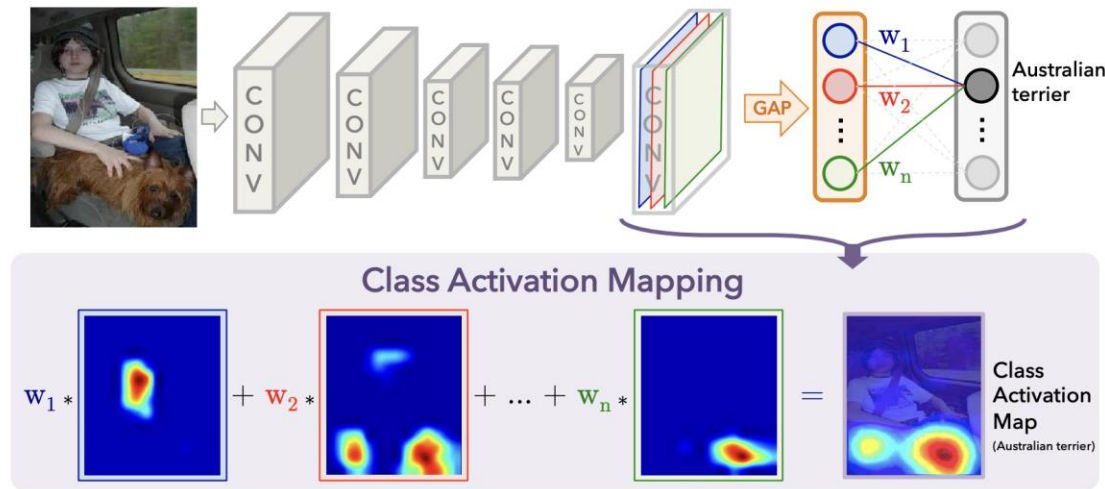
- Although σ may be difficult to tune, but this shows that including multimodal data could be beneficial

Interpretability

- One major critic of applying deep learning to materials problem
- Higher accuracy, less interpretable



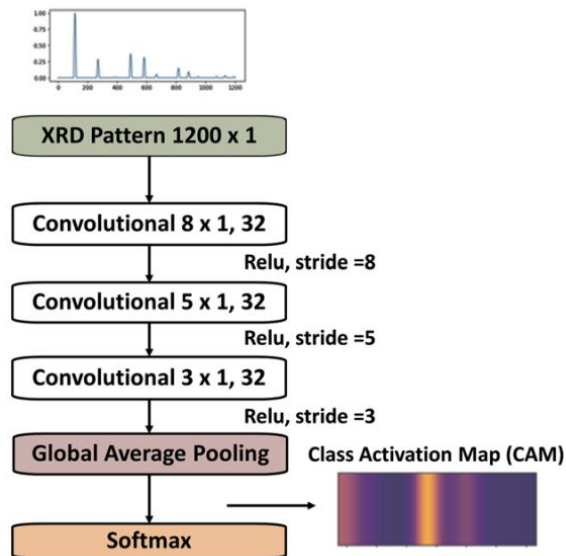
Class Activation Map (CAM)



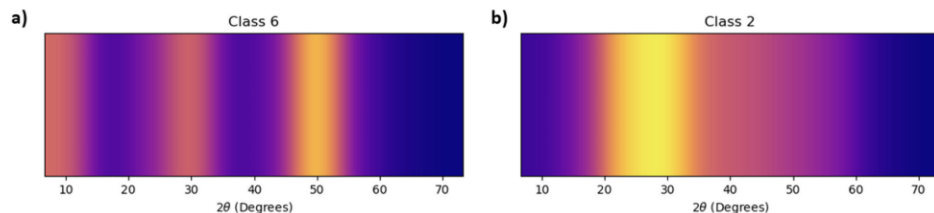
- After convolution and pooling layers, just put one fully connected layer to obtain the final classification
- Then the weight in that linear layer is the weight of each filter to the final classification
- Indicating where in the map lies the main discriminative features

Applying CAM

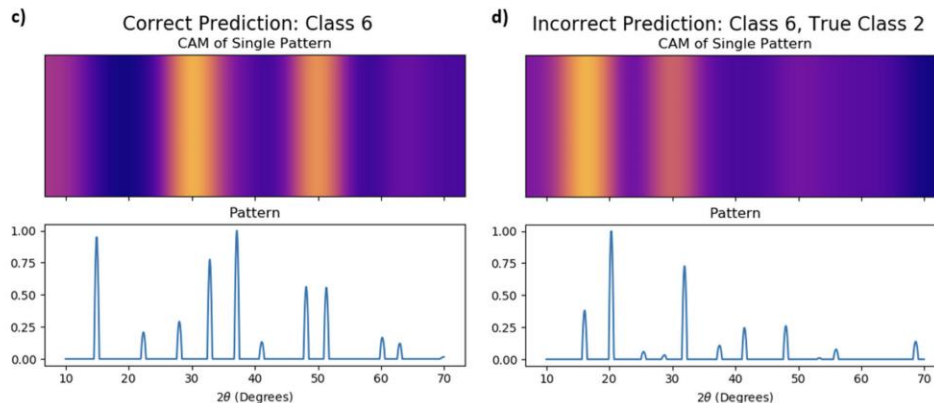
All Convolutional Neural Network



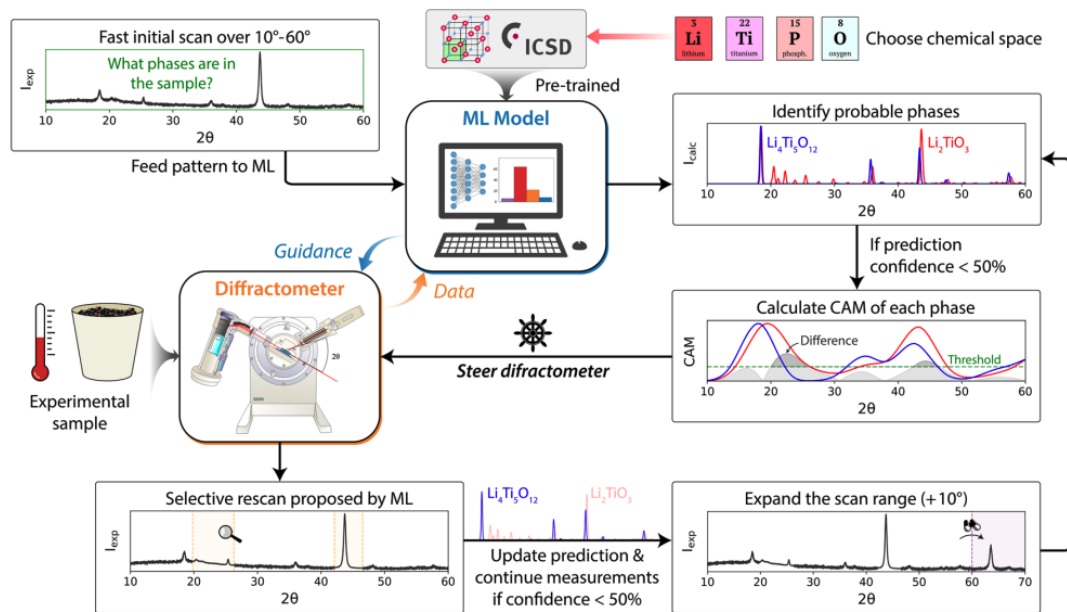
Average CAMs of certain phases



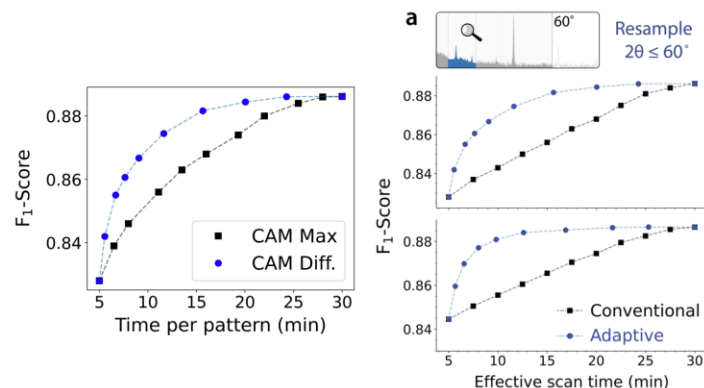
- Case with correct prediction gives similar CAM to the phase average CAM, and vice versa



Adaptive XRD Measurement

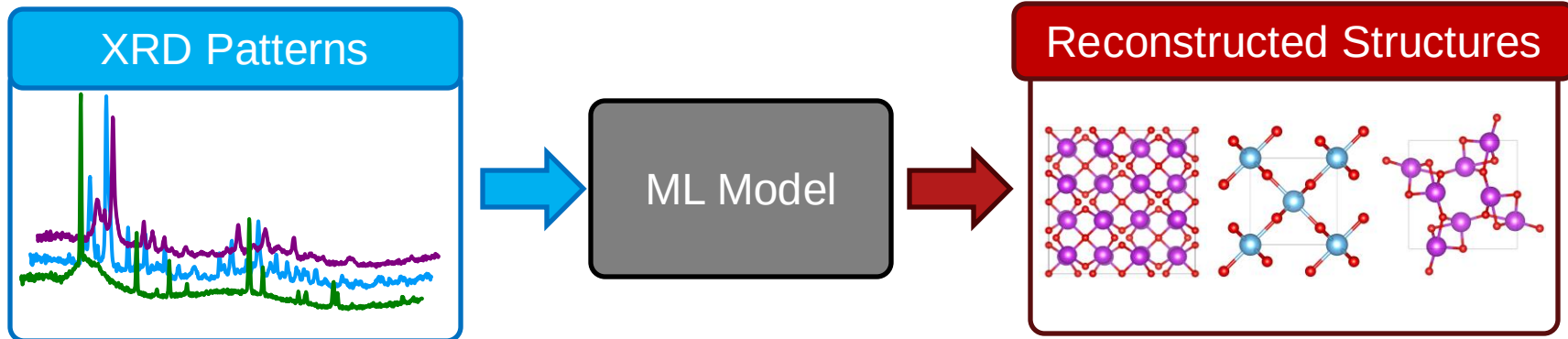


- Combine with probabilistic phase identification
- If not confident, rescan at regions that the top-2 phases' CAM deviate most
- Interpretability helps decision making down the line



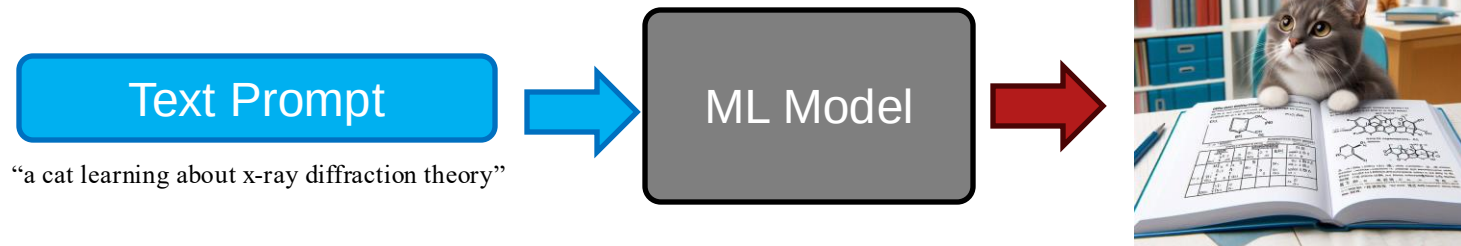
Can we go further?

- Given a XRD pattern, obtain crystal structure?



But these are completely different things!

Image generation

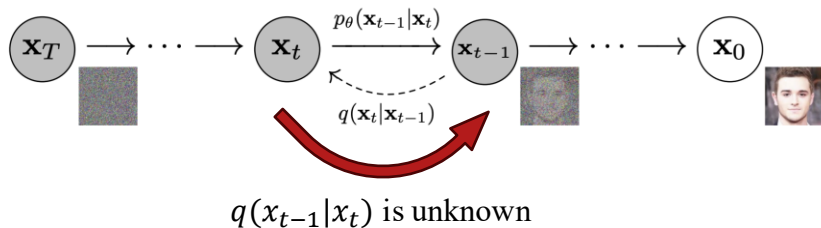


Diffusion Models

- A self-supervised method to learn a generative model

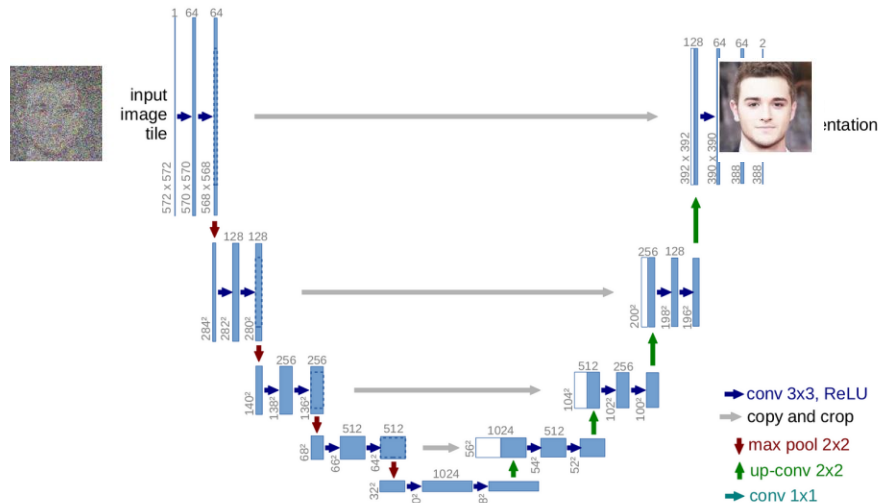
Forward process: Adding Gaussian Noise to data

Reverse process: Reconstruct image from noise



Q: Can we condition this model on an input?

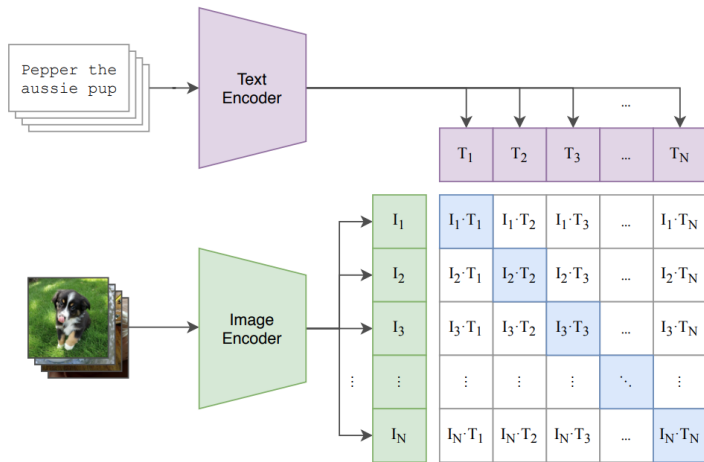
- The reverse process can be learned with a UNet-like structure
- Iterative denoising to obtain generated image



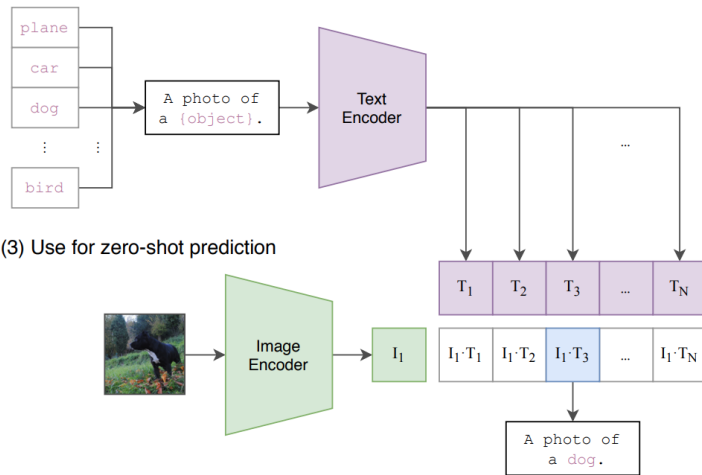
Align The Two Spaces – CLIP

Contrastive Learning Image Pre-training

(1) Contrastive pre-training



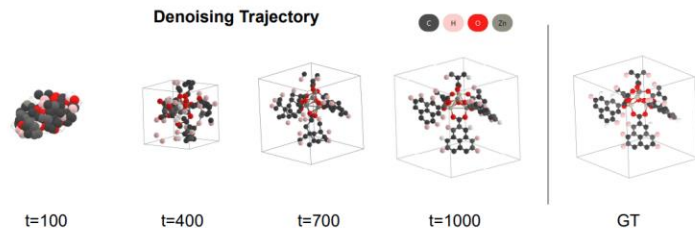
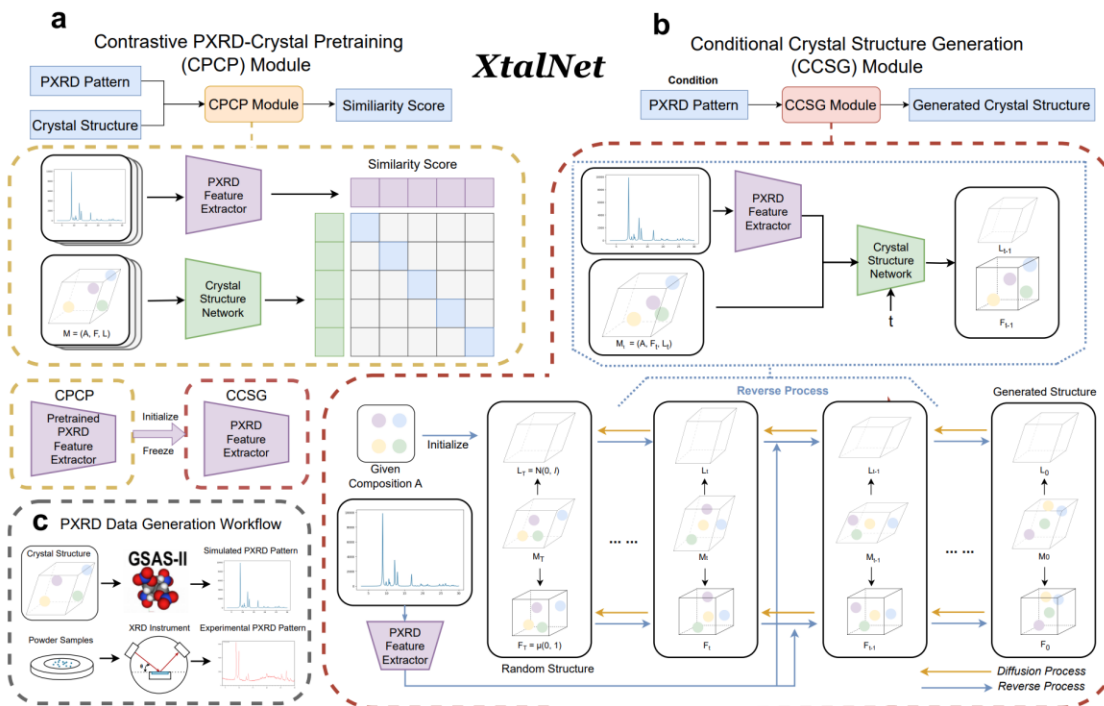
(2) Create dataset classifier from label text



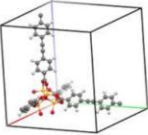
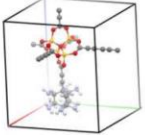
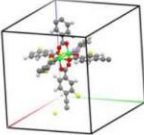
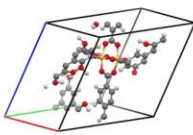
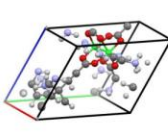
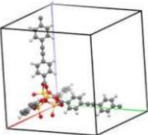
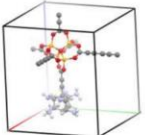
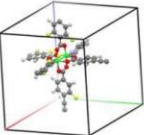
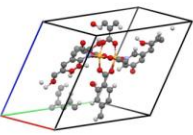
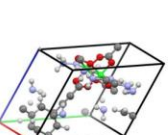
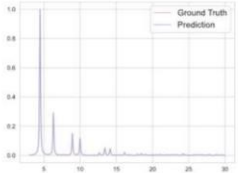
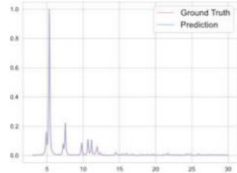
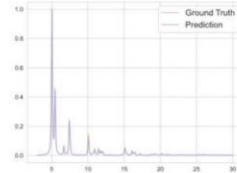
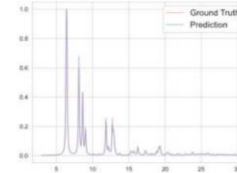
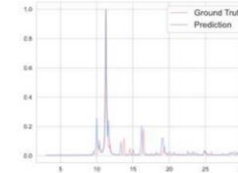
(3) Use for zero-shot prediction

- Train encoder pair and maximize the input pairs inner product in the encoding space
- Then you can have a single view for both space
 - Can be useful for transferring information from one to another

Direct Crystal Reconstruction with Diffusion Models



- Used transformer-based model and Equivariant graph neural network as encoders
- Append encoded XRD feature into the node of EGNN when doing diffusion on the generated crystal structure
- Did not mention whether it can deal with multiple phases

Formula	Zn ₄ H ₂₄ C ₄₈ O ₁₃	Zn ₄ H ₂₀ C ₃₀ N ₈ O ₁₃	Cu ₂ H ₁₆ C ₄₄ N ₂ O ₈ F ₈	Zn ₂ H ₂₀ C ₄₂ N ₂ O ₁₇	Cu ₂ H ₃ C ₂₂ N ₁₂ O ₈
Ground Truth					
Prediction					
RMSE	0.004	0.151	0.241	0.360	0.477
PXRD Pattern					
Cos Similarity	0.9994	0.9983	0.9982	0.9890	0.9090

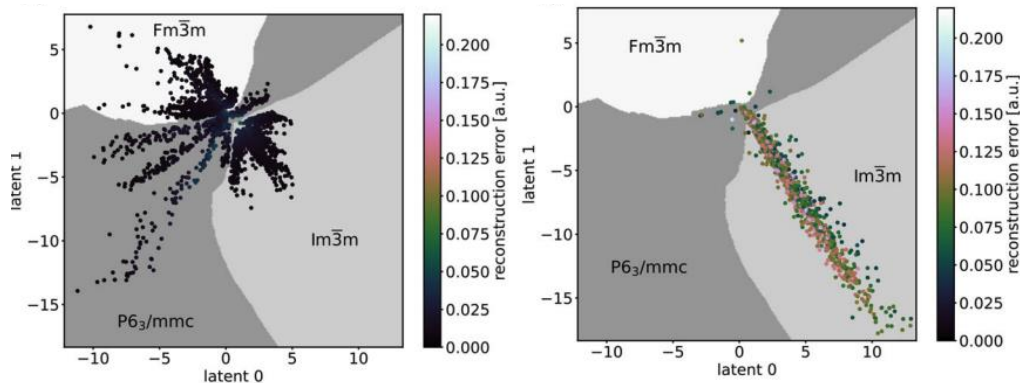
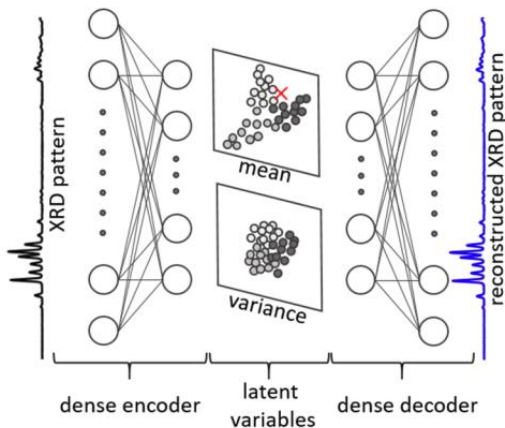


- High cosine similarity does not mean the XRD is close though

Great potential: If this can generate very large cell, may capture minute atomic disorder

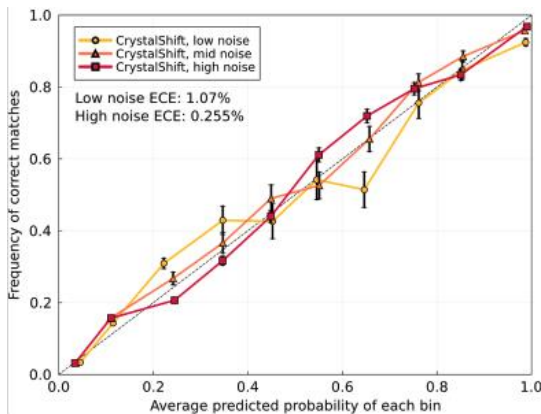
Underexplored: Novelty Detection

- What if the XRD pattern is not in the database or is just from a new crystal structure?
- One study on a very small scale: 3 CIFs
- Trained a variational autoencoder (VAE) with 5000 simulated pattern
- Threshold on reconstruction error to classify unknown phases
- Claim 99.16% accuracy on ideal, simulated data
- **Problem: Scalability? Mixtures?**

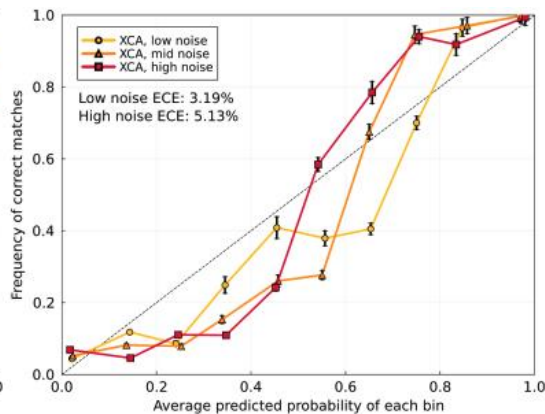
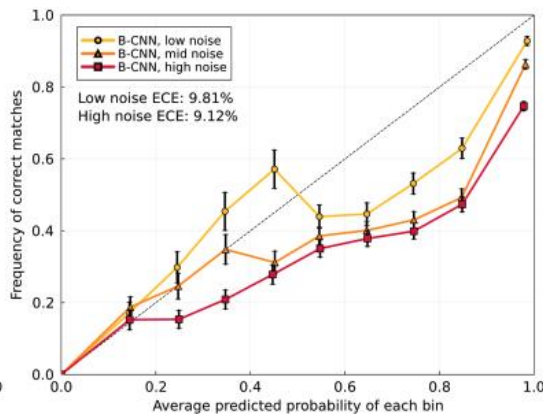


Underexplored: Probability Calibration

Bayesian-Based Method



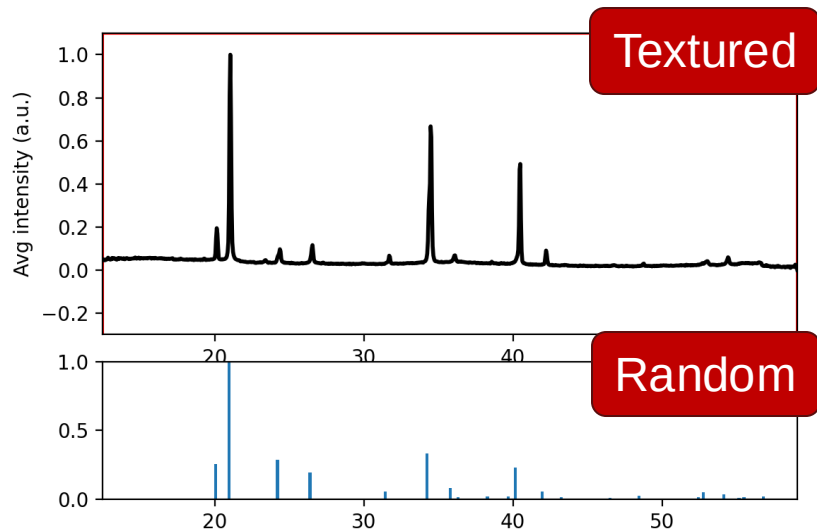
Probabilistic Deep Learning Methods



- Though different noise level data is included in the training data, prediction shift with noise level
- How to correctly, robustly train a calibrated DL model is something that needs to be study

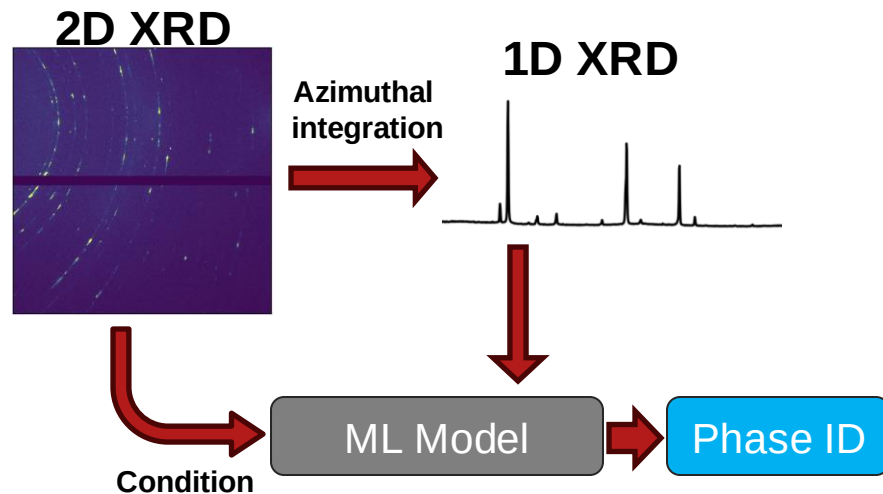
Futher reading: Guo, Chuan, et al. "On calibration of modern neural networks." *International conference on machine learning*. PMLR, 2017.

Underexplored: Texturing



- Peak intensity ratio changes
- Similar effect to atomic disorders
- Really difficult to deal with just using 1D

Introduce information from 2D XRD



- Information are lost once 2D XRD is integrated into 1D pattern
- Should be able to condition the ML model with 2D XRD and consider such information when considering the 1D XRD phase ID problem
- Consider relation between sets of peaks, e.g. (100) family in cubic

Problems: No Benchmark Dataset

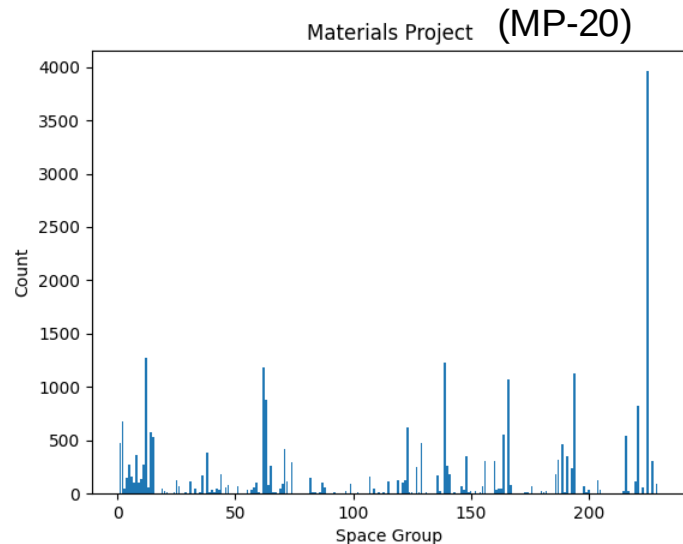
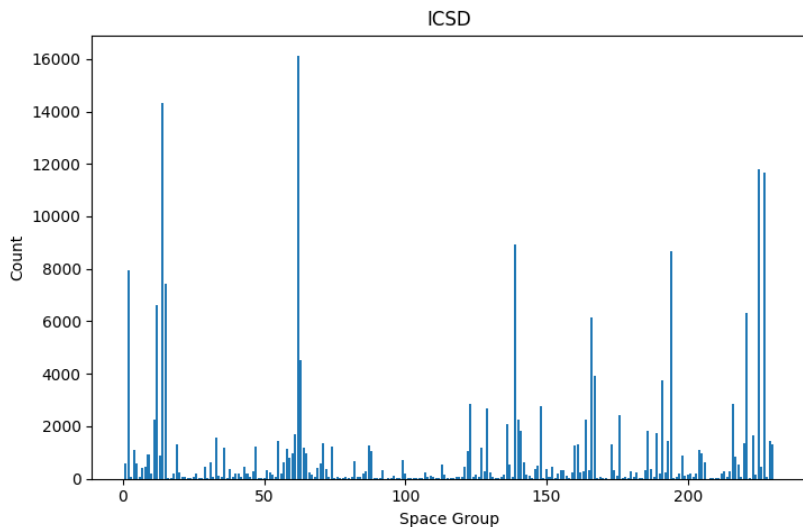
- Example: ImageNet
 - Computer vision field grows explosively after ImageNet is introduced
- Benchmark dataset provides a baseline for comparison between models
 - Allow research to focus on method/model development

Table 1: Summaries of existing powder XRD datasets. ICSD refers to the commercial Inorganic Crystal Structure Database. MP denotes the open-sourced Material Project.

Dataset	#XRD Pattern	#Structure	Open Access	Simulated	Crystal Source	Year
RRUFF [22]	3,002	3,002	✓	×	-	2015
XRDSP [55]	169,536	169,536	✓	✓	ICSD	2020
CNN [28]	1,785,405	170	×	✓	ICSD	2020
PQNet [14]	250,000	1	✓	✓	ICSD	2021
XRDAutoAnalyzer [57]	38,250	150	✓	✓	ICSD	2021
XRDIsAllYouNeed [26]	328,503	189,476&139,027 ³	×	✓	ICSD&MP	2022
AdvancedXRDAnalysis [27]	29,569,650	197,131	×	✓	ICSD	2023
CrySTINet [7]	100	100	✓	✓	ICSD	2024
CPICANN [5]	692,190	23,073	✓	✓	COD	2024
SimXRD	4,065,346	119,569	✓	✓	MP	2024

- A lot of variables in generating X-ray diffraction data
- It is hard to have one that covers enough circumstances

Problems: Real World is Skewed



- The data from ICSD and MP is highly uneven in space group
 - MP started by calculating materials for battery applications, a lot of entries contains Li

ICSD represent the distribution of real material structure

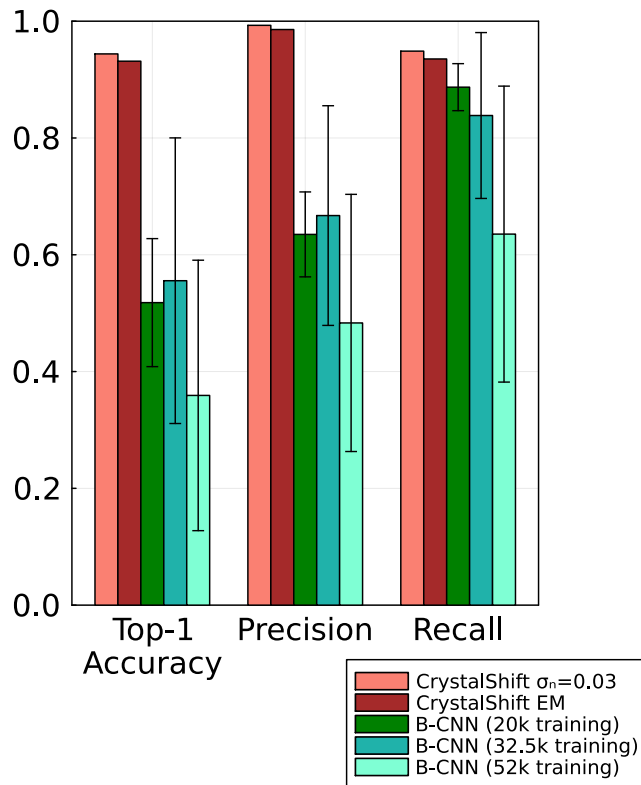
Phase ID should be independent of the underlying structure distribution

Problems: Inductive Bias

- Inductive bias is the set of assumptions that a learning algorithm makes to generalize from the training data to unseen data
- Locally the XRD pattern is not too interesting
 - They are just peaks
- CNN theoretically is harder to capture relations that are far apart, which is what contribute to a phase
- Future should move to methods with less inductive bias, e.g. attention-based models

Component	Entities	Relations	Rel. inductive bias	Invariance
Fully connected	Units	All-to-all	Weak	-
Convolutional	Grid elements	Local	Locality	Spatial translation
Recurrent	Timesteps	Sequential	Sequentiality	Time translation
Graph network	Nodes	Edges	Arbitrary	Node, edge permutations

Is Machine Learning Always Better?



- Compare to recently developed optimization and search based method, deep learning method did not out perform in some cases
- It suffers from
 1. Instability in training
 2. Overfitting to training sample
- We are still in the early stage of applying ML/DL for phase ID, use with caution

Final Takeaways

- Generally, it is still at the early stage of applying machine learning for phase identification
- Many newly-developed methods in the computer science community have potential in applying to this problem
- Future direction (in my personal perspective) are
 1. Better probabilistic multiphase labeling
 2. Novelty detection
 3. End-to-end crystal structure refinement
 4. Multimodal large language model

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